

Zhangir Nurmukhambetov

Engine and Tools Programmer

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PROFESSIONAL EXPERIENCE

Tools Programmer

April 2026 – Present

Flashgroup XR

Breda, Netherlands

- Developed a custom Unity Collider Editor that lets artists and programmers create physics colliders directly in the Scene View by selecting mesh vertices, previewing collider shapes, and generating boxes, spheres, capsules, cylinders, convex hulls, or CoACD-based convex decompositions. Iterated on the tool through artist feedback, simplifying the UI and improving workflow speed, visual feedback, undo support, and generated asset persistence.

Tools Programmer

November 2025 – April 2026

SportsImpro VR

Breda, Netherlands

- Delivered a full-body tracking prototype for Meta Quest by recreating VRChat's OSC-based tracking pipeline in Unity, enabling hip and leg movement to drive a VR tennis avatar. Researched existing tracking solutions, implemented an OSC receiver for Shiftall VR Manager data, and mapped tracker targets to a humanoid character using Unity IK.
- Built a Unity Editor one click video to FBX conversion pipeline by integrating MoveAI via a custom Python bridge, adding per clip progress and background execution, and enabling parallel conversions that cut a typical 3 video turnaround from about 15 minutes to about 5 minutes while removing manual upload, download, import steps.

PROJECTS

Custom Voxel-Based Engine Puzzle Platformer | C++, FMOD

May 2025 to Jun 2025

- Built a Unity style Particle System using EnTT ECS, cutting frame drops by 60%.
- Integrated 20+ FMOD sound events (e.g., footsteps, tongue, environment) for immersive feedback.
- Resolved 30+ gameplay bugs and led the Itch.io release, including trailer content and performance specs.

Custom Strategy Game Engine | C++, OpenGL, GLSL, EnTT, ImGui, Cereal

Feb 2025 to Apr 2025

- Built a real time terrain editor with brush tools and prefab support, used across 100% of game maps.
- Implemented GPU based terrain editing and tessellation rendering, reducing latency by 90%.
- Created JSON based scene serialisation with EnTT Meta, halving iteration time on level testing.

Custom Terrain Editor Tool | C++, OpenGL, GLSL, ImGui, GLM

Nov 2024 to Jan 2025

- Developed a GPU accelerated terrain sculpting tool with modular brush system and intuitive UI.
- Implemented heightmap deformation with dynamic normals and 80%+ reduction in visual latency.
- Designed foundational architecture reused in Fire Ant terrain editor, accelerating team ramp up and integration.

TECHNICAL SKILLS

Languages: C++, C#, GLSL, HLSL, Python

Frameworks & Libraries: OpenGL, EnTT (ECS), ImGui, Cereal, GLM, tinygtf, ImGuizmo

Game Engines: Unreal Engine 5, Unity (C#), Godot

Developer Tools: Visual Studio, Git, Perforce (P4V), GitHub Actions

Languages: English (Fluent), Russian (Fluent), Arabic (Beginner)

Interests: Creative writing, Digital art and design

EDUCATION

Breda University of Applied Sciences

Bachelor of Science in Creative Media and Game Technologies

September 2023 to Expected Jun 2026

Breda, Netherlands